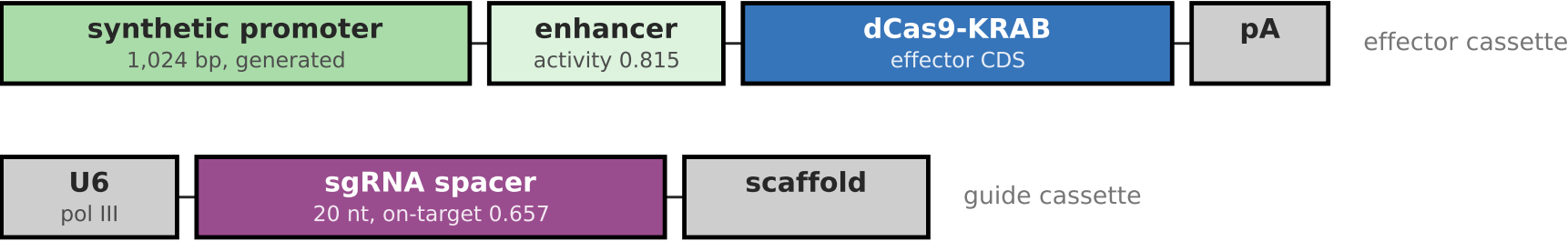


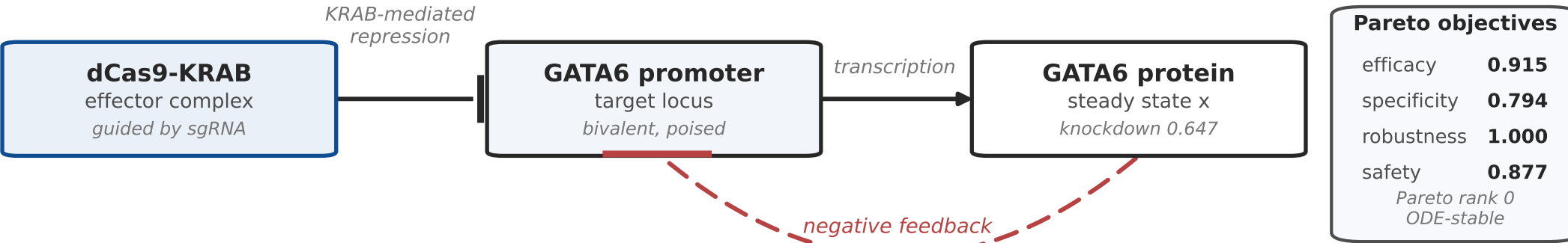
A designed CRISPRi circuit against GATA6, from construct to kinetics

a Delivered construct, two cassettes



Sequence-optimised for GC content, cryptic splice sites, restriction sites and codon adaptation

b Regulatory action and kinetics



$$\frac{dx_i}{dt} = \beta_i \prod_{a \in A_i} H^+(x_a) \prod_{r \in R_i} H^-(x_r) - \gamma_i x_i$$

$H^+(x) = x^n / (K^n + x^n)$, $H^-(x) = K^n / (K^n + x^n)$, β set by promoter strength, $n = 2$